The impact of manufacturing credentials on earnings and the probability of employment

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Abstract

This paper examines the labor market returns to earning industry-certified credentials in the manufacturing sector. Specifically, we are interested in estimating the impact of a manufacturing credential on wages, probability of employment, and probability of employment specifically in the manufacturing sector post credential attainment. We link students who earned manufacturing credentials to their enrollment and completion records, and then further link them to their IRS tax records for earnings and employment (Form W2 and 1040) and to the American Community Survey and decennial census for demographic information. We present earnings trajectories for workers with credentials by type of credential, industry of employment, age, race and ethnicity, gender, and state. To obtain a more causal estimate of the impact of a credential on earnings, we implement a coarsened exact matching strategy to compare outcomes between otherwise similar people with and without a manufacturing credential. We find that the attainment of a manufacturing industry credential is associated with higher earnings and a higher likelihood of labor market participation when we compare attainers to a group of non-attainers who are otherwise similar.

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Introduction

One of the main pathways to higher wages for those entering the workforce is the attainment of an education beyond high school completion. Over the past several decades, the wage gap between high school graduates and those with a college degree has widened (Autor, 2014). At the same time, however, the nominal cost of a four-year degree has become increasingly expensive, and the economic return to certain majors and their associated skill sets may not compensate for a heavy debt burden (Goldrick-Rab, 2016). In an environment where students need post-secondary skill attainment to succeed but cannot afford college, many sub-baccalaureate options have appeared to fill the gap. One such pathway, which we study in this paper, is industry credentialing.

In collaboration with the National Association of Manufactures (NAM), the Manufacturing Institute, and the National Student Clearinghouse (NSC), the U.S. Census Bureau acquired data on industry credential attainers and their education records, which were then linked to demographic and wage data as part of a pilot to study this type of educational pathway. An industry credential is, in essence, a certificate of skill attainment that is issued by an industry organization upon the formal demonstration of the skill. These unique data allow us to examine patterns of credential attainment by student characteristics and to evaluate the impact of attainment on labor market outcomes. This paper covers the results from our pilot investigation, where we focused specifically on manufacturing credentials issued by two separate industry organizations.

We find that the attainment of a manufacturing industry credential is associated with higher earnings and a higher likelihood of labor market participation when we compare attainers to a group of non-attainers who are otherwise similar. Labor market gains are concentrated

among attainers who go on to work in manufacturing, although we also find some spillover effects in other industries. In examining the credential-attainer population alone, we find that wage gains are sustained over a five-year window of earnings records; that more credentials—especially "stackable" credentials—are associated with higher earnings; and that attainers in different parts of the labor market life course see wage gains post attainment, including older workers experiencing stagnant earnings prior to attainment. We also examine the timing of credential attainment vis-à-vis the timing of traditional degree attainment, finding that post-credential attainment of an associate degree leads to even higher earnings. These findings provide evidence on how stepwise skill attainment may provide a viable pathway for labor-market success.

Our paper dives into the topic of industry credentials, providing an overview on how they differ from or are similar to other types of sub-baccalaureate education. We then provide a series of summary analysis that are meant to paint a picture of the credential population and tease out some patterns of attainment by the demographic and labor market characteristics of students.

Background and Literature

1. *What do we mean by an industry credential?*

As used in this paper, an "industry credential" is a formal documentation by an industry organization of a student's skill attainment. Specifically, a credential is "An occupational certification . . . a credential awarded by an authoritative body—such as an industry or professional association—to an individual who demonstrates designated knowledge, skills, and abilities in a particular occupation."[2] Our definition differentiates such industry credentials from

[2] Counting-Credentials-2021.pdf (credentialengine.org)

other types of skill attainment, including degrees, educational certificates, and occupational licenses. However, most industry credentials implicitly or explicitly require educational prerequisites before the final test or skill demonstration, tying credentials to the education system to ensure candidates' abilities. Industry credential requirements provide specific guidance on which competencies are needed to perform different jobs. Because of these narrow guidelines, "learners understand and document their own knowledge, skills and abilities relative to labor market requirements, thereby increasing their competitive advantage, qualifying for new and evolving jobs in labor markets, and advancing in their careers over time."[3] Established competency standards make credentials transferable across employers and connect the acquired skill base clearly to other credentials, certificates, and degrees.

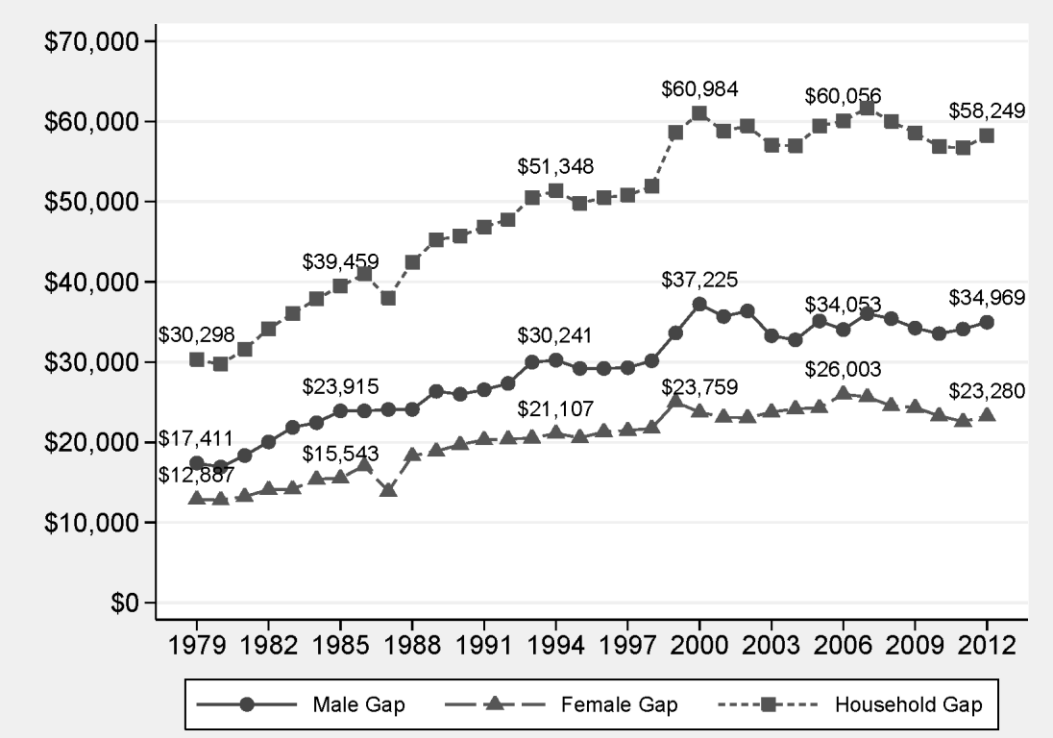
While "certificates, certifications, and licenses are often lumped together into one category reserved for credentials other than the academic degrees,"[4] industry credentials such as those we study differ from other categories of credentials on several metrics. First, unlike a degree or other education credential, industry credentials are occasionally time-limited (this is not the case, however, for the credentials we examine in this paper). In this manner, they more closely resemble occupational licenses but are awarded by an industry organization rather than a government agency. Industry credentials bear similarities to degrees in that they are often embedded into programs offered by education and training providers, including educational institutions, but the real distinguishing characteristic of them is that they are designed and awarded either by an association or a standards-setting organization made up of industry representatives, or they are awarded by an employer or equipment vendor in the industry. A credential is earned via a "written, oral, or performance-based assessment."[2]

[3] Quality Dimensions for Connected Credentials, The American Council on Education, 2016
[4] Building Better Degrees Using Industry Certifications (newamerica.org)

Credential attainment also bears closer resemblance to degree attainment in that they may be "stackable," where a new credential builds on the skills attained at a lower level. A credential attainer may receive training for a few months to a year to achieve a level of skill that gains them an entry-level job, and then attain additional skills as needed for advancement and higher earnings. Such a trajectory has become increasingly common in certain fields, including health care and information technology (RAND Corporation, 2021). In our context, we focus on industry credentials awarded by two manufacturing organizations. According to the data provider, these credentials represent approximately 30 percent of the available vendor-neutral credentials awarded within the manufacturing industry each year. Each organization provides credentials for a variety of skills within the industry, as well as levels of certification (e.g., Machining I, II, and III) that allow us to assess the impact of stackable credential attainment.

The U.S. has seen an increase in the provision and attainment of credentials such as these (Kim & Tamborinin, 2019) as it has experienced decreasing job-market returns to educational attainment below the baccalaureate level. Figure 1, from Autor (2014), shows how the ratio in earnings between workers with a high school diploma versus those with a bachelor's degree has changed over the decades. When the high cost of a four-year degree is taken into account, the lifetime return on investment in a bachelor's remains high (about 14 percent), although it has decreased somewhat since its peak in the 1990s (Abel & Deitz, 2019). Sub-baccelaureate education has garnered attention from government, industry, and non-profit entities as a way to close this gap in eduational returns at a lower cost than a four-year degree.

Figure 1: The ratio in earnings between workers with a high school diploma versus those with a bachelor's degree



Source: Reproduced from Autor (2014) with permission from the author.

2. Previous research on sub-baccalaureate attainment generally/credentials specifically

Although a wide literature exists on the labor market returns to education (Burgess, 2016), less is known about the impact of industry credentials, separate from other types of educational attainment, on labor market outcomes. A general consensus is that, conditional on pre-attainment earnings and demographic characteristics, those who demonstrate skill through some formal recognition have greater long-term success in the labor market, whether that skill was acquired via sub-baccalaureate education or licensing (Baird, Bozick, & Zaber, 2021; Kleiner & Krueger, 2013; Minaya & Scott-Clayton, 2020; Jepsen, Troske, & Coomes, 2014; Gittleman, Klee, & Kleiner, 2018).[5] Research suggests three main pathways by which attainment contributes to

[5] See Belfield & Bailey (2017) for a review of related literature.

success: the process of attainment itself (Maguire, Freely, Clymer, Conway, & Schwartz, 2010); the signaling component to credential attainment, where the credential itself gives employers information about a potential employee's skill (Blair & Chung, 2018); and a gatekeeping element that controls labor supply through formal testing and thus increases wages (Kleiner & Krueger, 2013).[6]

Although the popularity of non-degree credentials has increased (Duke-Benfield, Wilson, Kaleba, & Leventoff, 2019), data collection regarding non-degree educational attainment has not kept pace until recently. Additionally, the type of credential we examine has not been easily separable from other types for the purposes of analysis. That being said, much of the research into sub-baccalaureate credential attainment in general—of which there is a substantial library—should be informative about the returns to industry credentials specifically.

This "substantial library" has, for the most part, combined all sub-baccalaureate attainment into an aggregate "some college" category (Kim & Tamborinin, 2019; Tamborini, Kim, & Sakamoto, 2015), in spite of the wide variation in the types of schools and programs reflected.[7] Industry credentials, in particular, may be attained through community college classes, a proprietary training program, or employer-sponsored training. The development and accreditation of such training programs often result from coordination between training providers, employers, and the industry association backing the credential and applying its requirements (Holzer, 2015). In light of the way in which these programs have evolved, some attention has been spent measuring the impact of such "sectoral" training[8] separate from the

[6] In the realm of licensing, federal- and state-level requirements, including testing, create a barrier to entry for non-licensed individuals in a way that local or other non-binding credentials do not. Such barriers suppress the number of workers available for an occupation and drive wages up (Kleiner & Krueger, 2013).
[7] An examination of public-use ACS income data from 2019 reveals that workers with "some college" have higher income dispersion, expressed via 90-10 ratios, than do workers of other education types, providing some evidence on the variability of returns to different kinds of sub-baccalaureate attainment.
[8] "Sectoral" in that the "training is well-targeted towards the needs of employers" (Holzer, 2015).

attainment itself; these studies (e.g. Maguire, Freely, Clymer, Conway, & Schwartz, 2010) find that attendees of sectoral training programs see a two-year earnings improvement of as much as 30 percent over non-attendees.

Meanwhile, an active area of research examines the impact of community or technical college certificates, up to an associate degree, on labor market outcomes through linking administrative enrollment records (often a single state) to earnings records. A recent example is Stevens, Kurlaender, & Grosz (2019), who find wage returns that average 14 percent for certificates and 45 percent for associate degrees, depending on the field of study. Similarly, Minaya & Scott-Clayton (2020) find a positive impact on earnings of long-term certificate attainment and show a stronger positive and more long-lasting impact for associate-degree attainers. Although such attainment bears more resemblance to a degree rather than a credential, these studies are relevant through the technical nature of the curriculum and the types of students whom these programs attract. A limitation of these administrative-records-based studies is an absence of demographic information beyond gender and a restriction to earnings records from within the same state (neither of which is a limitation in our case).[9]

More specific recent research into credentials, licensing, and other types of skill demonstration has led to new variables being added to surveys conducted by the Census Bureau in support of Federal agencies.[10] These new variables resulted from the work of the Interagency Working Group on Expanded Measures of Enrollment and Attainment (GEMEnA), a group tasked with developing questions that would capture participation in and attainment of credentials and licenses. Previous to the work of this group, survey responses to credential or license attainment suffered from poor response rates (Kleiner & Krueger, 2013) or a mismatch

[9] For more information on the impact of attrition on measures of state-level earnings, see Foote & Strange (2019).
[10] An example is the Current Population Survey.

between measures of attainment and questions regarding education (Baird, Bozick, & Zaber, 2021). Thus, before the availability of the GEMEnA data, research relied on imputing the presence of an occupational license based on state or local law and comparing workers within an industry across states with and without licensing requirements (see, for example, Redbird, 2017).

Using these new data, Baird et al. (2021), found causal evidence that the attainment of an industry credential leads to increased employment and higher earnings, especially among the sub-baccalaureate population. These findings provide evidence on the role of industry credentials, specifically, in providing higher wages among those who never obtain a bachelor's degree. Baird et al. (2021) relies on an instrumental variables identification strategy: the proportion of individuals in the same locality and demographic group "stand in" for the presence of credential.

Our analysis provides improvements over research based on survey responses or combined college and earnings records. First, the addition of administrative records on credential attainment, earnings, and income should improve precision over survey values. Second, the unique nature of our data allows us to account not only for the presence of a credential, but the type and number of credentials and the timing of their attainment. The longitudinal nature of our data allows for a close examination of the wage trajectory of credential earners over a decade of labor-market participation. Finally, the addition of demographic characteristics from decennial and survey data allows us to provide details on how effects vary by characteristics other than gender (e.g., race, Hispanic origin, household characteristics and geography).

Research questions

Our research questions expand on what is already known about the impact of sub-baccalaureate education but orient the inquiry specifically to credentials earned from an industry organization. We examine the impact these credentials have on the short- and medium-term employment trajectories of attainers, focusing first on summary impacts by different demographic characteristics within the group of attainers. We then ask how earnings and employment trajectories for these attainers compare to a subset of workers who are observationally similar on many measures, but who did not attain a credential.

Because we look to a representative sample of U.S. working age adults to create our control group of workers who do not attain a credential, we are assuming that this group did not attain one of these manufacturing credentials (because they did not link to the project data) *and* that they do not have a comparable credential from a similar organization. While it is easy to determine the first condition, we cannot confirm the second: our industry credential providers may issue approximately 30 percent of credentials in manufacturing, but this percentage is difficult to estimate and changes regularly due to the increasing popularity of these types of certifications. That being said, to the extent that our "control" group might contain members with similar credentials, our estimates should be considered, first, a lower bound and, second, an effect on attainers of these specific credentials and not credentials more generally.

Data

1. *NAM-NSC*

The main data used in this project stem from an arrangement between NAM and NSC whereby individuals receiving industry credentials from four separate organizations were linked

to their enrollment and education records.[11] The time frame for credential achievement spans 2005 to 2018, while degree attainment and other education variables span a much wider period (the earliest year of degree is the late 1960s). We received credential data from four organizations, two of which are in general manufacturing (the third is in health care and the fourth is in welding).

Variables included in the combined NAM-NSC data are the number and type of each credential, as well as credential categories and levels; highest degree attained and year of attainment; program of degree study; and personal information such as name and date of birth. We start with about 120,000 individuals who have attained a credential.

2. PIK assignment

For each dataset linked in the project, the Census Bureau assigns a unique, time-invariant person identifier by comparing personally identifiable information (PII) in the dataset to similar information in a master reference file. This identifier, called a Person Identification Key (PIK), is nine digits and randomly assigned. PII comprises social security numbers, names, dates of birth, and address information such as place of birth or address history. When a dataset contains an SSN, such as tax data, PIK placement approaches 99 percent. For Census-produced survey data, PIK placement rates are lower (85 percent to 90 percent), and they vary based on socio-demographic characteristics. Once a PIK is placed on a file, dataset observations can then be

[11] For the purposes of the Family Educational Rights and Privacy Act (FERPA), the NSC data that we are using in our project is "directory information," which entails a subset of the information that NSC collects about students. "Directory information" is information contained in the education record of students that would not generally be considered harmful or an invasion of privacy, if disclosed, and as identified by an educational institution, it typically includes information such as name, address, telephone listing, date, and place of birth, major or field of study, enrollment status, dates of attendees, and degrees, honors, and awards received. If an education institution defines any of those fields as non-directory, NSC will not provide that data to Census. NSC may share or use directory data for research purposes without student consent under FERPA. Note directory data is based upon a universe/data capture mechanism that allows students to opt out of sharing their information.

matched to their information in any other demographic or administrative dataset in which they also appear.

For this project, names and, for some observations, dates of birth were available in the NAM-NSC data, leading to a PIK rate of 56 percent. While considerably lower than survey data PIK rates, this rate is not out of range for third-party-provided data that has similarly limited PII.[12] Table 1 provides details on the rate of PIK placement and how well our project observations with a PIK linked to other data sources.

Table 1. Match rates of project data to unique identifier, demographic data, and federal tax information

PIK rate on full NSC-NAM data transfer	0.5766
PIK rate on manufacturing credential records	0.5628
Fraction unique manufacturing learner IDs assigned PIK	0.5433
Percent of PIKed manufacturing cred students linked to ACS/Decennial	0.8203
Percent of PIKed manufacturing cred students ever linked to W2	0.8281
Percent of PIKed manufacturing cred students ever linked to 1040	0.8767

Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data.
Approved for release under CBDRB-FY2021-CES010-029.

3. Demographic data

The next step involved matching the PIKed observations in the NAM-NSC data to their appearance in the 2010 decennial census or any American Community Service data from 2005 to 2017. We were able to link approximately 82 percent of the PIKed students who get a manufacturing credential to demographic data in the combined decennial and ACS, allowing us to assign race, ethnicity, sex, and age to these cases. For observations that linked to both decennial and ACS, we took the last-reported value for any demographic variable.

[12] PIK rates tend to be better for adults who have interacted with the labor market, filed taxes, or participated in other economic behavior. PIK rates may be especially low for our project observations because as a group they tend to be younger and less "connected" (see Bond, Brown, Luque, & O'Hara (2015)).

As mentioned previously, our population-level coverage of manufacturing credentials should comprise about 30 percent of credential-holders in manufacturing. However, because of the PIK placement rate, we may miss some matches between the demographic data and the list of credential-holders, which again may weaken the association between credential attainment and employment outcomes due to credential-holders in our "control" group. We discuss the implications of mismatch in our results section.

4. Tax data

The next part of the data build involved linking the NAM-NSC data to federal tax information from Form W-2, Form 1040, and the Business Register (BR). From the W-2, we captured annual earnings information for students based on their five highest-paying employers. We also captured the employer identification number (EIN) for the highest-paying employer for each year of the W-2. W-2 information is available from 2005 to 2018. The link between the W-2 data/1040 data and the NAM-NSC students was especially high, with 83 percent of students having W-2 information for at least one year and 88 percent linked to at least one year of 1040 data.

From the most-recent version of the BR, we match two-digit North American Industry Classification System (NAICS) industry identifiers[13], total employees, and total payroll based on the EIN of the highest-paying employer. We can examine year-by-year changes in their employers' two-digit NAICS code for students to assess whether they are working in their field of credential.

[13] https://www.census.gov/naics/

From Form 1040, we link the wages and salary field, the filing status field, and fields for the number of children claimed as dependents. The wage and salary field on Form 1040 is less useful than W-2 earnings for two reasons: first, a spouse's earnings may be included for married filers; second, earnings and salary may include other elements of earned income besides wages from an employer or self-employment. Filing status and children claimed are important measures of family structure that we use to supplement information we link from the next data set, the American Community Survey (ACS).

5. ACS 2016 to 2018

We made a further link to the ACS for 2016-2018 for two purposes. First, we find observations from the NAM-NSC-tax data who also appear in an ACS from years 2016 to 2018 *by matching on PIK* and compare further demographic and labor market information across datasets. Second, we select from the ACS observations a set of working-age adults *not matched on PIK* to the project data who can be used in our coarsened exact matching strategy. These unmatched observations are linked to W-2, 1040, and BR information to capture the same earnings and industry variables described earlier. We focus on linking these later years of data (2016 to 2018) to capture earnings for NAM-NSC observations and non-credential holders in years after most credential-holders have attained at least one credential.

The ACS data provide information on educational attainment, household structure, income and earnings, and self-reported industry classification. To develop our initial comparison group from the ACS, we require that observations be of prime working age (18 to 65) and that they do not appear in the NAM-NSC data. As mentioned previously, there is a possibility that these unmatched observations have received a credential from a different organization.

6. Variable definitions

These disparate data sets provide overlapping information on several measures. For example, the NAM-NSC data has detailed information on degree attainment, while the ACS has information on years of education. Table 2 reports each variable used in our analysis and how we derived the variable. The data set listed first is the primary source for the variable, and it is supplemented with information from the second-listed dataset if it is missing in the first.

Table 2. Variable sources and definitions

Variable	Type	Source(s)
Credential	0/1 indicator (cred)	Observation appears in project data
Credential type	Category	Project data
Credential level	Category	Project data
Credential date	Date	Project data
Age	Number	Project data/ACS
Education	4 categories	Project data (degree attainment); ACS (years converted to degree category)
Family structure	4 categories	IRS 1040/ACS (combination of married/single and presence of children)
Race/ethnicity	4 categories	ACS/Decennial (missing race category included for non-matches)
Gender	0/1 indicator (male)	ACS/ Decennial /Project data
Census region	4 categories	IRS 1040/IRS W-2/Project data (ed institution)
Industry	5 categories	IRS W-2/Business Register/ACS (NAICS codes)
Employer payroll	Number	IRS W-2/Business Register
Earnings	Number	IRS W-2/ACS

Graphical comparisons

Our first set of analyses focuses on examining labor market outcomes of project observations vis-à-vis the timing of credential attainment and demographic characteristics. The goal of these analyses was to uncover patterns in credential attainment and subsequent outcomes by types of students and workers; these patterns ultimately became informative for the development of our matching strategy and regression modeling.

For our graphical comparisons, we identify the date of each observation's last credential attainment, thus capturing the time at which a worker earned his or her highest formal skill set

before entering into or continuing in the labor market. We trace out average earnings for different demographic and industry groups five years before and five years after the last credential. These analyses collapse real earnings by time and by time-invariant and -variant groups, such that when an observation can change a category (say, industry of work) over time, he or she is included in the category for that year-by-type cell. Earnings are inflated to 2018 dollars using the January Consumer Price Index for All Urban Consumers from the Bureau of Labor Statistics.

We also provide graphs showing state-level patterns in employment and earnings pre- and post-manufacturing credential for the top ten states with the highest concentration of manufacturing credentials. These include California, Florida, Illinois, Indiana, North Carolina, Ohio, Pennsylvania, South Carolina, Texas, and Virginia. Geographic information is obtained using the residential address reported on income tax returns and may differ for an individual across years.

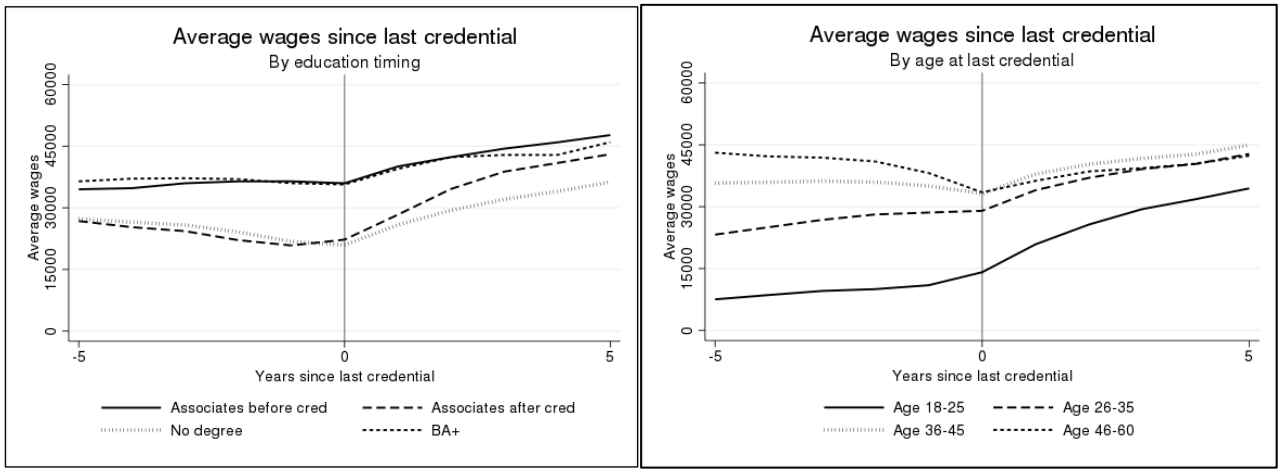


Figure 2a and 2b. Average wages since last credential by education timing and age group. Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY2020-CES014-001.

The overwhelming majority of credential attainers are high-school graduates with less education than a four-year degree. Industry organizations provide manufacturing credentials as a pathway to better skills, and thus higher earnings, for this population. In Figure 2a, we show the

patterns of earnings for those who earned a credential by the ultimate level of education they attain—in particular, those without a post-secondary degree, those who earned an associate degree before earning the manufacturing credential, and those who earned an associate degree after their credential. We can see that both the degree and credential may independently confer earnings benefits, although the higher pre-credential earnings for those who already had an associate degree is partially a result of differences in age at the time the credential was earned—very few 19-year-olds likely already had an associate degree.

Next, we investigate the extent to which patterns in earnings vary by age group. Although the majority of students who earn a manufacturing credential are between ages 18 and 35, there remain many workers pursuing credentials late in their careers. Figure 2b shows that receiving a manufacturing credential is associated with a steeper upward trend in earnings for all age groups. Earnings exhibit a dip or a flattening in the period before credential receipt. This may be due to credential receivers spending unpaid hours earning the credential. Noticeable in this graph is the pre-credential downward wage trajectory for earners ages 46-60—credential receipt may allow older workers to staunch a downward earnings trend.

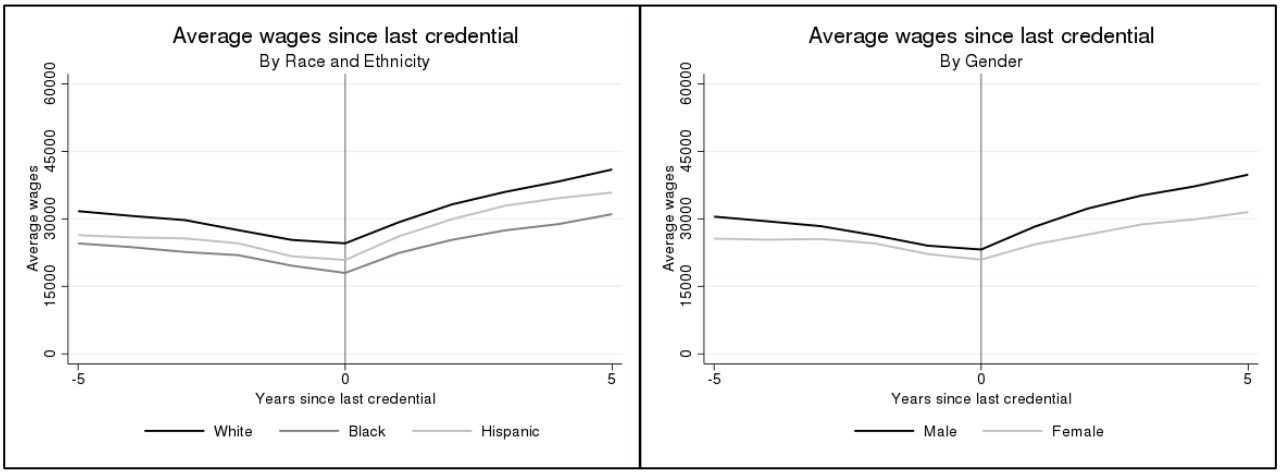
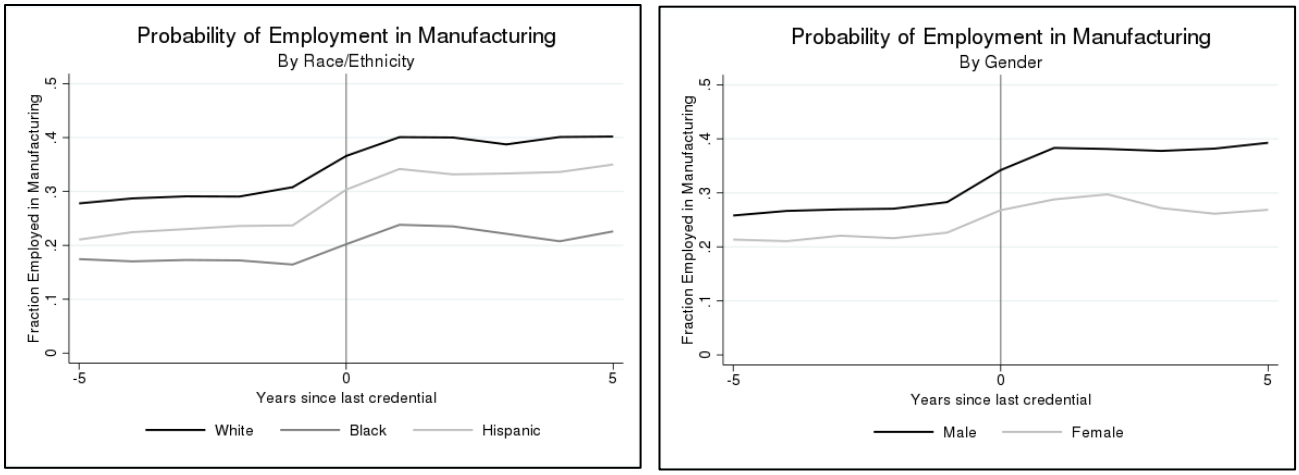


Figure 3a and 3b. Average wages since last credential by race/ethnicity and gender. Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY2020-CES014-001.

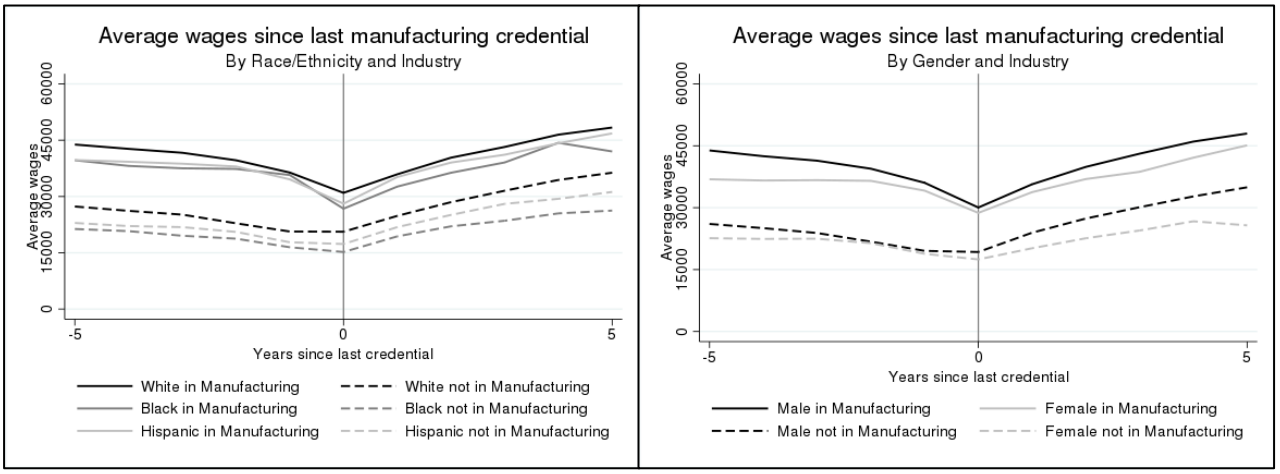
Another goal in promoting the acquisition of manufacturing credentials by workers is to reduce earnings inequality across gender and racial/ethnic groups by having clearly defined requirements. Despite this, in Figure 3a we see differences on average across racial-ethnic groups, with white workers having the highest earnings both before and after obtaining a credential, and Black workers having the lowest. A similar pattern is reflected in Figure 3b, with women having lower earnings relative to men both before and after earning a manufacturing credential.

These differences in average earnings across groups could arise for a variety of reasons, but most importantly if there are differences in the probability of employment in the manufacturing sector. In Figure 4a, we see that although all workers experience a jump in the probability of employment in the manufacturing sector after earning a credential, there are differences by race/ethnicity and gender. About 40 percent of white workers with a credential are employed in manufacturing in contrast with 23 percent of Black workers and 34 percent of Hispanic workers. However, when we split earnings by race/ethnicity for those employed in manufacturing and those employed in another sector, we can see in Figure 5a that the gaps in



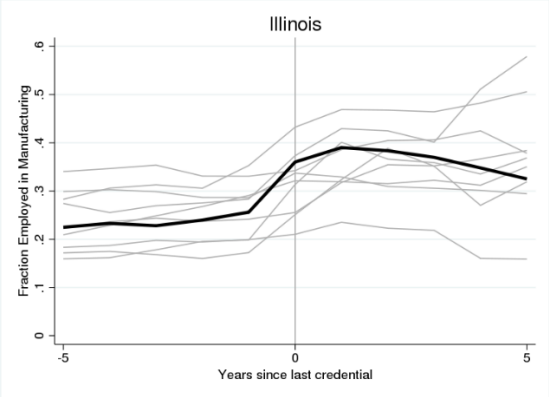
Figures 4a and 4b. Probability of employment by race/ethnicity and gender. Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY2020-CES014-001.

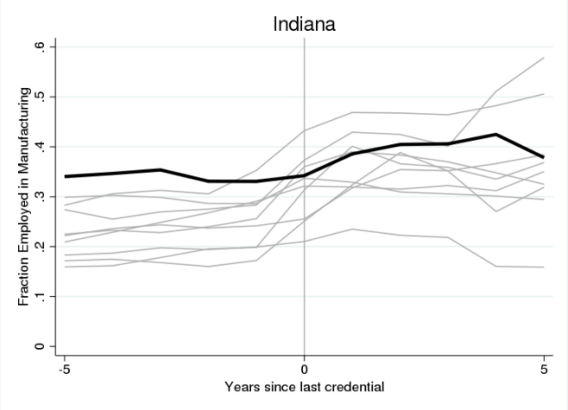
earnings by race/ethnicity are much narrower post-credential for those employed in manufacturing. This story also holds true for women versus men, as shown in Figure 5b. These findings suggest that credentials may even the playing field somewhat—in other words, there may be unrealized earnings gains for Black, Hispanic, and female workers if they were paired more effectively with employers in the manufacturing sector.

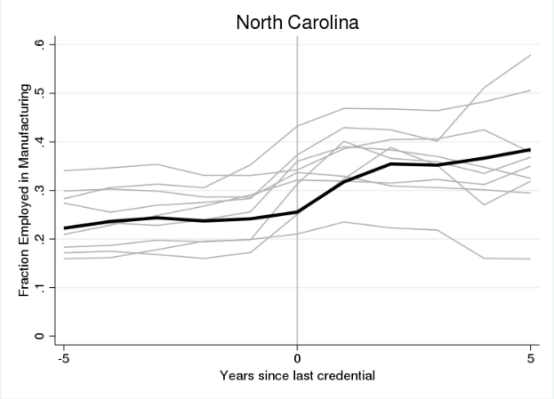


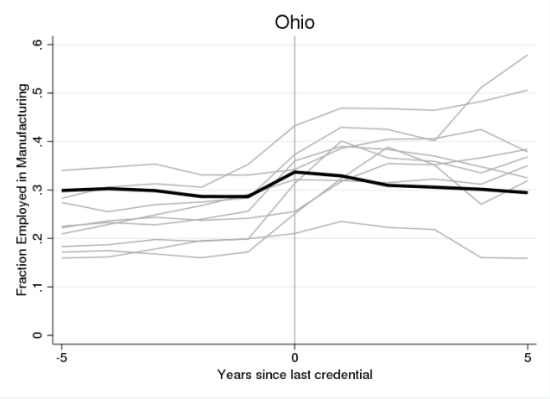
Figures 5a and 5b. Average wages by race/ethnicity and gender by industry. Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY2020-CES014-001.

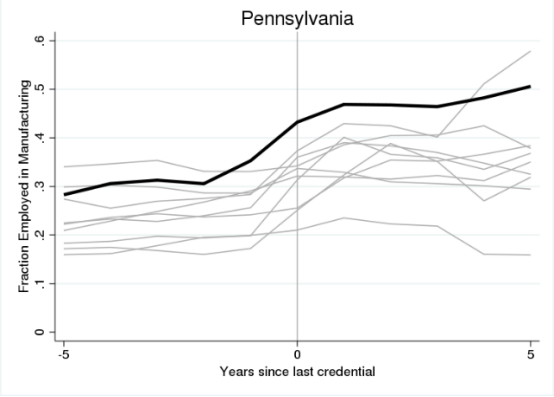
Finally, the returns on credentials may differ significantly across locations, with some states having more favorable labor market conditions for those with manufacturing credentials reflected both in average earnings and in probability of employment in the manufacturing sector. We present earnings and employment patterns for the top ten states in number of credentials awarded. Figure 6 indicates large discrepancies across states in the probability of employment in the manufacturing sector after a worker obtained their last credential, with Pennsylvania and South Carolina experiencing the largest gains and Texas seeing minimal gains. Florida, by contrast, sees the largest increase in earnings post-credential. The appendix that accompanies this paper shows state-by-state wage differences.

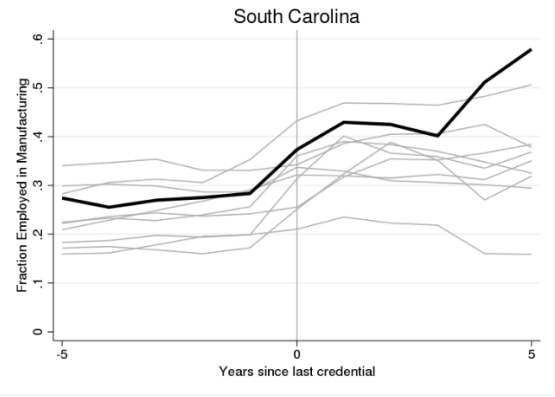

Illinois
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Years since last credential

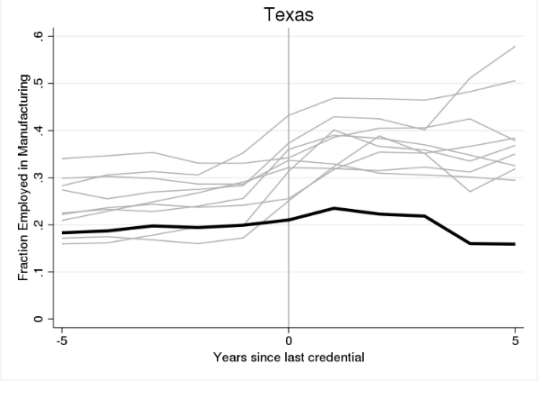

Indiana
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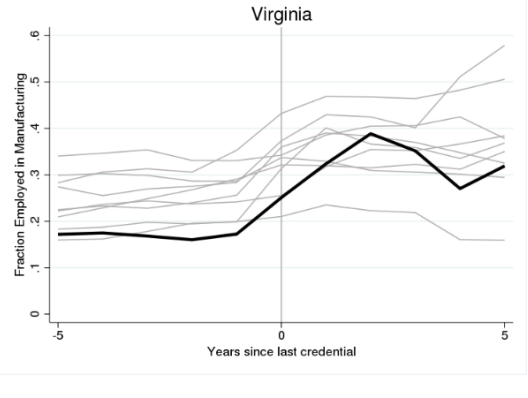

North Carolina
Fraction Employed in Manufacturing
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Years since last credential


Ohio
Fraction Employed in Manufacturing
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Years since last credential


Pennsylvania
Fraction Employed in Manufacturing
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Years since last credential


South Carolina
Fraction Employed in Manufacturing
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Years since last credential


Texas
Fraction Employed in Manufacturing
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Years since last credential


Virginia
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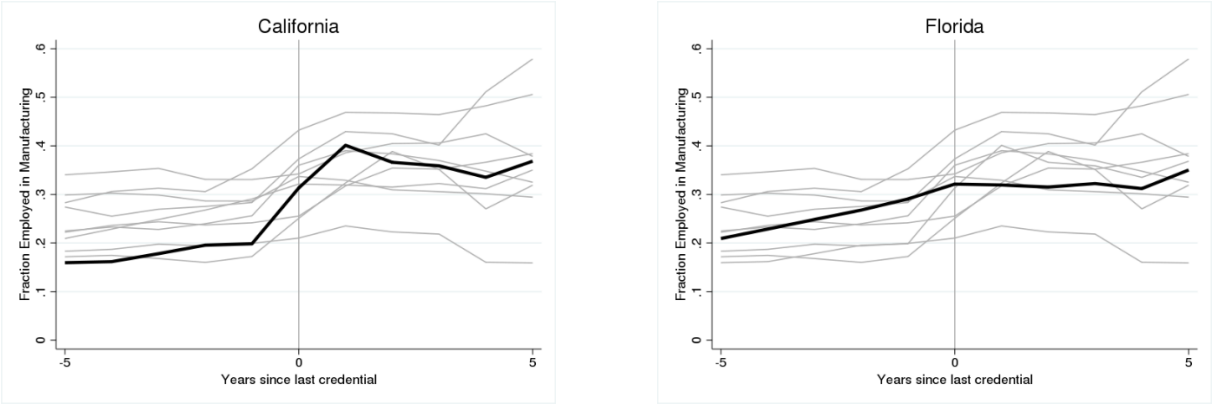


Figure 6. Probability of employment in manufacturing by state. Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY2020-CES014-001.

Matching analysis

1. Regression model

We generate a control group using the ACS observations from 2016 to 2018 who did not appear in the project data, restricting the sample to adults between the ages of 18 and 65 so that the universe of observations is working age. Our regression results are based on canonical models of labor-force outcomes, where earnings are a function of demographic, family, and industry characteristics. Specifically, we model the simple regression

$$y_i = \alpha + X_i * \beta + \varepsilon_i$$

where y_i is, in turn, 2018 earnings or 2018 employment (0/1) and X_i includes the following: a quadratic in age; gender; four race/ethnicity categories (non-Hispanic White, non-Hispanic Black, non-Hispanic other race, and Hispanic); four family-type categories (marital status by children); region of employment; employed in 2014; firm payroll in 2017; and 5 quantiles of "pre-period" earnings averaged between years 2014- 2016.

2. Description of CEM as identification strategy

The preceding OLS model suffers from bias in that credential earners select into the type of education they choose; thus credential earners are likely to be "different" from non-credential earners in ways that a traditional regression analysis cannot control for. We attempt to correct for this selection bias using a matching strategy that compares control observations who "look like" project observations on pretreatment observables. While this method does not eliminate selection bias, it balances control and treatment observations in a way that allows for more plausible comparisons between the two groups.

Matching strategies are common in the economics literature as a way to replicate, as closely as possible, a randomized experiment in which treatment and control units are balanced with respect to their pre-treatment characteristics. The steps in a matching include: a definition of "closeness" used to determine goodness of match between a treated and a control unit; implementing the matching algorithm based on that measure; performing assessments of the match and refining if needed; and proceeding with final estimates of the treatment effect given the final matching choice (Stuart, 2010). While exact matching is the ideal scenario, in practice it is unattainable in high-dimensional data: increasing the number of cells on which an exact match must occur means an unacceptable loss of sample and thus increased bias in the estimates.

Coarsened exact matching (CEM), which we employ here, allows for a wider possibility of matches by increasing the bin width of matching variables. For example, rather than match exactly on age, the CEM process allows for matching with bins of ages (start of career, prime working age, and near-retirement, in our case). Upon creating the coarsened dataset, X^*, from the original data X, every observation from X is assigned to a stratum defined by a unique observation of X^*. Strata with at least one control and one treatment unit are retained. Many of

our choices on coarsening are driven by the summary analysis presented earlier; for example, we assume differences in expected outcomes to be driven by an observation's race and gender and what industry he or she works in. Our final binning choices balance these assumptions with the goal of retaining as many treated observations as possible.

Once the CEM process is completed, a weighted regression analysis proceeds using a weight for each observation (strata with unmatched units receive 0 weight; treated observations have a weight of 1; and matched controls receive a weight that is proportional to their stratum assignment). As our final analysis, we restrict project observations to those who attained a credential in 2016 or 2017, allowing us to perform the equivalent of a matched difference-in-differences analysis, where we establish clear pre- and post-periods.

Results

1. *OLS*

Table 3a. Unweighted OLS Results, Earnings

	Full sample (1)	Associate or less (2)	Associate or less, cred 2016-2017 (3)
Has credential	-5,643	1,146	1,502
	169.3	89.61	143.6
Observations	8,323,000	5,569,000	5,541,000
R-squared	0.065	0.143	0.142

Table 3b. Unweighted OLS Results, Probability of Employment

	Full sample (1)	Associate or less (2)	Associate or less, cred 2016-2017 (3)
Has credential	0.0028	0.0015	0.0149
	(0.0013)	(0.0014)	(0.0022)
Observations	8,323,000	5,569,000	5,541,000
R-squared	0.5908	0.5891	0.5895

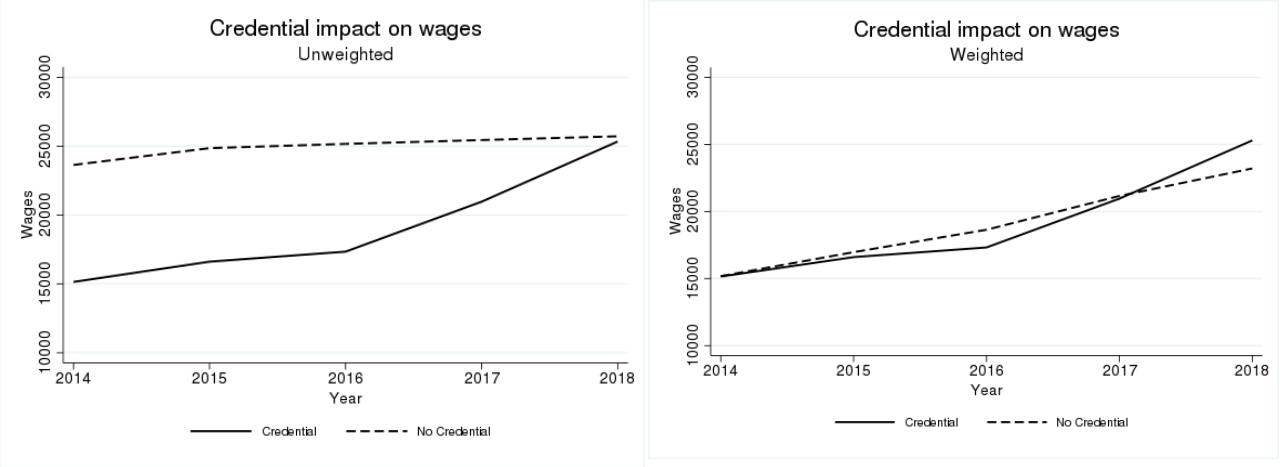
Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY22-CES014-003.

To benchmark our matching estimates, we first report unweighted OLS regression results for the impact of earning a manufacturing credential on earnings and employment for three increasingly restrictive samples: (1) the full sample of people ages 18 to 65; (2) people ages 18 to 65 with an associate degree or less; and (3) people ages 18 to 65, with an associate degree or less, who obtained their last credential between 2016 and 2017. In the full sample, having a credential is associated with -$5,643 in 2018 earnings. However, it is clear from the next model that these results depend on the inclusion of those with more than a bachelor's degree. We can see in Table 3 that imposing the sample restriction that individuals have attained an associate degree or less is important: for this group, having a credential is associated with $1,146 *more* in earnings. Few people in our project data have a bachelor's degree, and we are essentially allowing those with higher education in the ACS to drive the results in model 1. Model 3 further limits the sample to those who receive their credential between 2016 and 2017, giving the results a difference-in-differences interpretation. Here we see an even stronger association between receiving a credential and wage, with credential holders earning $1,502 more in 2018.

2. Matching estimates

A concern with the OLS estimates is that credential attainers may select into attainment due to low earnings, and this selection biases the OLS estimates upward in the sub-baccalaureate group. In other words, our credential attainers may have unobserved characteristics, such as stronger ambition, that we need to account for in our estimates. To make the samples more comparable, we implement coarsened exact matching, which matches on characteristics in the treatment group (who have manufacturing credential) and the control group (ACS respondents who do not have a credential). A key test of whether matching is successful relies on comparing

the "pre-period" outcome for the "treatment" and "control" observations. In Figures 7a and 7b, we can see that the matching algorithm does a good job of reweighting observations so that the pre-period earnings in model 3 for the non-credentialed group match the pre-period earnings of the group receiving a credential between 2016 and 2017. There is a slight disagreement remaining in the period immediately before credential attainment when the treatment observations appear to experience a dip in earnings due to hours spent in training or education.



Figures 7a and 7b. Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY2022-CES010-001.

Table 4 reports the results of the matching strategy using our 3 sample restrictions. We find that when we match and run our regressions against the full sample of ACS workers, credential attainment is still associated with a lower 2018 wage—about $1,165 less in 2018 earnings. As mentioned, the attainment of a bachelor's degree is more common among the ACS workers than among the credential holders. When we limit the sample to those who have an associate degree or less, we find that credential attainment leads to about $1,468 more in 2018 earnings. Finally, when we restrict both samples to having an associate degree or less and to comparing only those

who have clear pre- and post-periods, we see that the attainment of a credential is associated with about $1,993 more in annual earnings.[14]

Table 4. Coarsened exact matching results: Earnings

	Full sample (1)	Associate or less (2)	Associate or less, cred 2016-2017 (3)
Has credential	-1,165	1,468	1,993
	88.46	83.68	128.3
Observations	6,410,000	4,140,000	3,138,000
R-squared	0.3956	0.3775	0.2509

Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. approved for release under CBDRB-FY2022-CES010-001.

Table 5. Coarsened exact matching results:, Probability of Employment

	Full sample (1)	Associate or less (2)	Associate or less, cred 2016-2017 (3)
Has credential	0.0062	0.0069	0.013
	0.0013	0.0013	0.0022
Observations	6,410,000	4,140,000	3,138,000
R-squared	0.4066	0.4048	0.3658

Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY2022-CES010-001.

Next, we look at the probability of employment in 2018 in our three specifications (Table 5). We find that those who have a credential are more likely to be employed regardless of how we define the sample. The impact is economically small, however, ranging between six-tenths of a percentage point to one percentage point. This suggests that the benefit of earning a credential operates more strongly through the type of job a student obtains, rather than as a process for matching them to a job at all.

To better understand the mechanisms underlying the earnings results, we repeat the matching procedure separately for workers employed in five main sectors of industry: Manufacturing; Administrative Support and Waste Management; Transportation and Warehousing; Wholesale or

[14] Results are robust to excluding from the control group all ACS respondents whose education was imputed.

Retail; and Other Industry or Missing Industry. These industry categorizations comprised the top five industries where observations from our project data worked post-credential. We are interested in whether the manufacturing credential boosts earnings because it provides skills relevant to the manufacturing sector or if the act of attaining a credential confers spillover benefits, such as connecting students to a network of employers, revealing some underlying worker characteristic to employers, or generally signaling achievement.

Table 6. Coarsened Exact Matching Results, Earnings by Industry

		Full sample	Associate or less	Associate or less, cred 2016-2017
Industry				
Manufacturing	Has credential	1,845	4,693	5,405
		159.2	149.6	237.9
	Observations	536,000	408,000	352,000
Administrative Support and Waste Management	Has credential	-1,456	458.5	701.6
		227.1	216.6	327.3
	Observations	293,000	217,000	150,000
Transportation and Warehousing	Has credential	888.8	2507	1,338
		379.3	382	558.3
	Observations	216,000	171,000	99,000
Wholesale or Retail	Has credential	1,270	3,302	3,530
		283	261.5	420.3
	Observations	416,000	309,000	210,000
Other Industry or Missing	Has credential	-3,741	-903.1	-121.8
		125.8	115.7	183.4
	Observations	5,080,000	3,109,000	2,397,000

Source: NAM-NSC, decennial 2010, and 2005-2018 ACS, Form 1040, and Form W-2 data. Approved for release under CBDRB-FY2022-CES010-001.

We can see in Table 6 that in our preferred specification, a manufacturing credential leads to higher earnings in all four of the major industry categories, but we cannot reject the null for workers in other sectors or for those who are missing an industry code (potentially due to

unemployment). Earnings gains are largest for those who work in manufacturing—manufacturing workers who obtained a credential make on average $5,405 more than manufacturing workers without a credential. The second largest earnings bump is for credentialed workers in the Wholesale or Retail industries.

There are two explanations for why we might see such strong positive results for workers outside of the manufacturing sector. The first is that the credentialing programs may offer a benefit to students beyond the skills contained in course material, or that the attainment of the credential sends a signal of general productivity to the employer. The second is that we are not able to effectively compare workers by occupation, and our definition of industry may be leaky—i.e., there may be occupations that directly benefit from skill attainment in manufacturing within industries such as warehousing, retail, and transportation.

Despite the positive evidence that obtaining a manufacturing credential is associated with increases in employment and earnings, in this paper we are unable to disentangle human capital improvement from signaling effects. That is to say, these results could be driven by specific skills students acquired through attending the training sessions, or they may provide employers with information about workers' higher productivity—an attribute that may have led them to enroll in a credential program in the first place.

Conclusion

In this paper, we examined a specific type of sub-baccalaureate educational attainment—an industry credential. This study was motivated by new research interest in pathways to living wages beyond the attainment of a four-year degree.

We find that workers who attain an industry credential in manufacturing have higher earnings and are more likely to be employed than workers who are similar on many metrics but who do not attain a credential. In our preferred specification, a worker with a credential can expect to earn around $2,000 more per year than a comparable worker without a credential. Because we are looking at manufacturing credentials specifically, it should come as no surprise that we observe the largest wage gain upon comparing these two types of workers within the manufacturing industry. However, we also provide evidence that credentials may provide spillover skill, or at least spillover signaling, that confers benefits in several different industries.

These findings result from a pilot project between the National Association of Manufacturers, the Manufacturing Institute, the National Student Clearinghouse, and the U.S. Census Bureau. Future expansions to this project will lead to multiple opportunities to study the impact of different types of credentials on labor market outcomes. Future work will incorporate new credentials from additional data providers and will take a deeper dive into the connection between the impact of credentials and characteristics of the labor market, such as growth in the manufacturing industry, local differences in unionization rates, and the "stackability" of credentials. We expect this research to expand in tandem with the number and variety of credentials being offered to U.S. workers.

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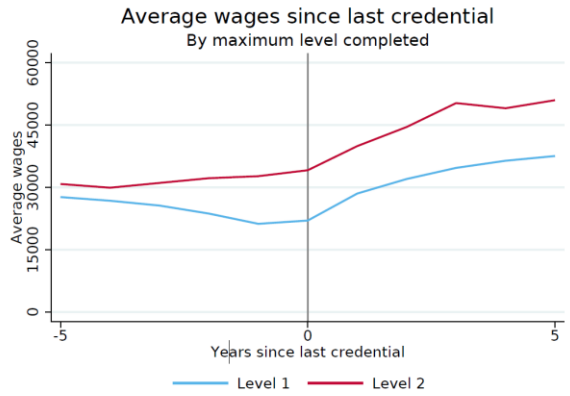
Stevens, A. H., Kurlaender, M., & Grosz, M. (2019). Career technical education and labor market outcomes evidence from California community colleges. *Journal of Human Resources, 54*(4), 986-1036.

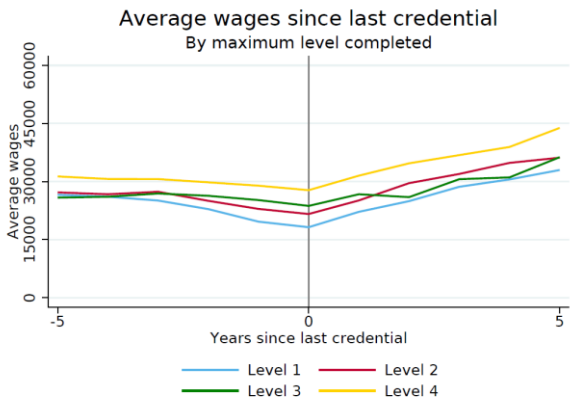
Tamborini, C. S., Kim, C., & Sakamoto, A. (2015). Education and lifetime earnings in the United States. *Demography, 52*(4), 1383-1407.

Appendix to The impact of manufacturing credentials on earnings and the probability of employment[1]

A. Stackable credentials

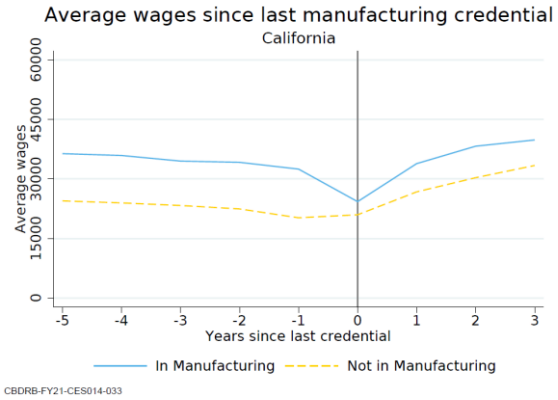
The manufacturing credentials we examine in the paper have "stackability"—meaning that a student can attain additional credentials for a given skill that build on previous credentials. In this manner a series of credentials can resemble degree attainment, where more complex topics build on foundation courses in a particular subject. We find that conditional on attaining any credential, additional credentials are associated with higher earnings.

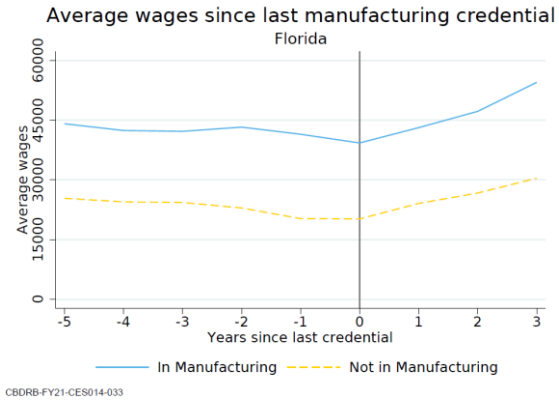




B. Top 10 credential holder states

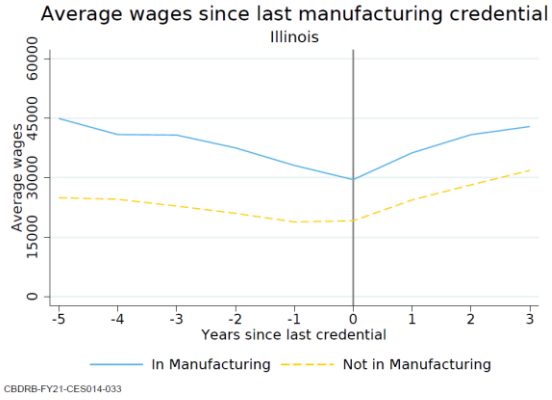
Our manufacturing credential data covers workers in the United States, but several states stand out as having a large number of credential holders from the two manufacturing organizations. The graphs that follow show state-specific information on the return to credential holding for those who work in manufacturing versus those who do not.

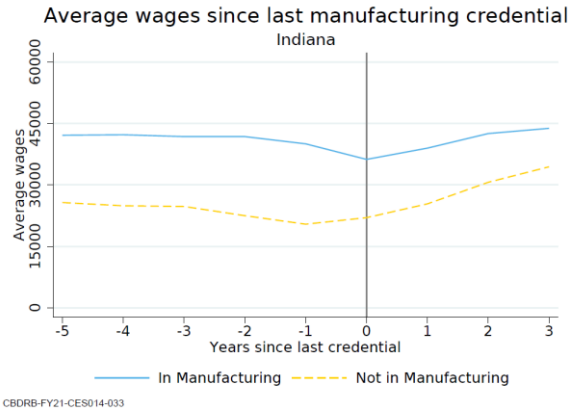


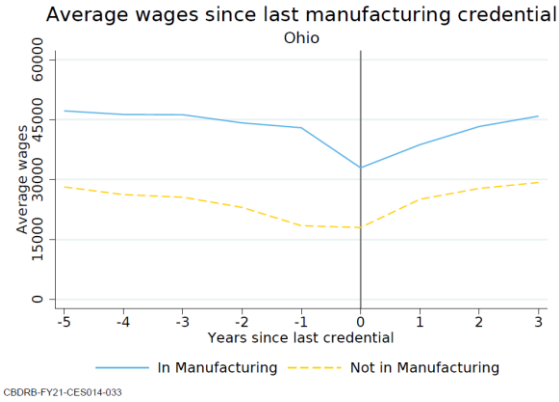


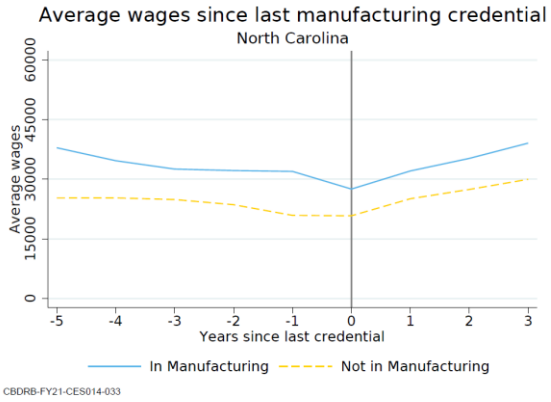
[1] Sources: NAM-NSC data, decennial 2010, and 2005-2017 ACS, Form 1040, and Form W-2.

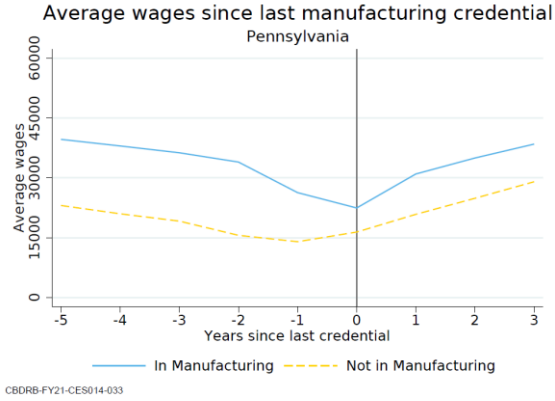
In all cases, credential holders who work in the manufacturing sector have earnings that are as high or higher than credential holders who work outside of the sector. We find that earnings are highest in Florida for those who work in manufacturing, and that the gap between those who work in manufacturing and those who do not diverges the most in this state. The gap is smallest in Virginia.

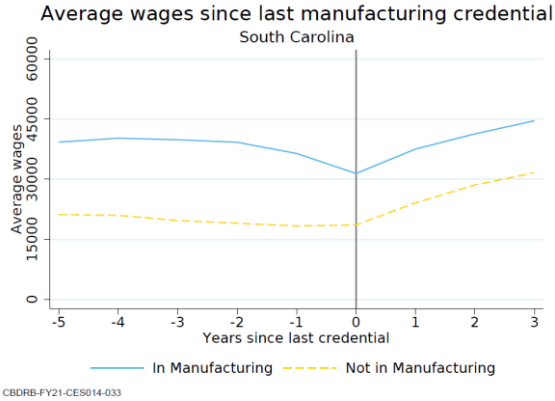


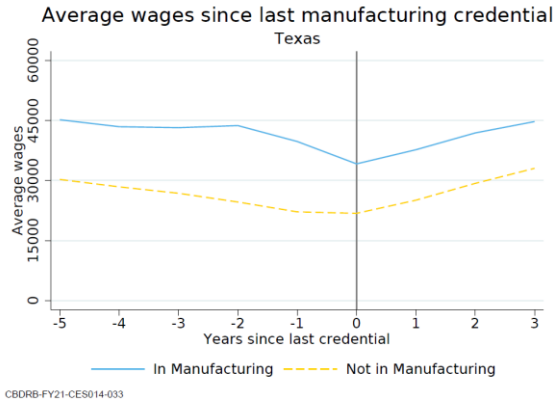


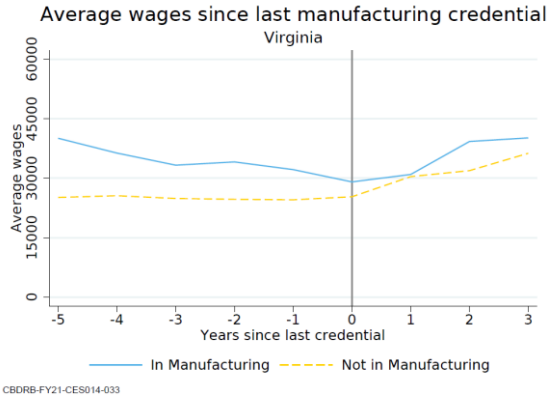









Average wages since last manufacturing credential
Texas
Average wages
0
15000
30000
45000
60000
-5
-4
-3
-2
-1
0
1
2
3
Years since last credential
In Manufacturing
Not in Manufacturing
CBDRB-FY21-CES014-033


Average wages since last manufacturing credential
Virginia
Average wages
0
15000
30000
45000
60000
-5
-4
-3
-2
-1
0
1
2
3
Years since last credential
In Manufacturing
Not in Manufacturing
CBDRB-FY21-CES014-033